The progress of this version of the UCB policy is shown in Figure 5.10. This time, UCB is able to find the global optimum thanks to the higher level of exploration induced by the larger value of β. However, if β is so large such that UCB only spends its budget exploring the search space, then our optimization performance might also suffer. (We will see an example of this in this chapter's exercises later on.)

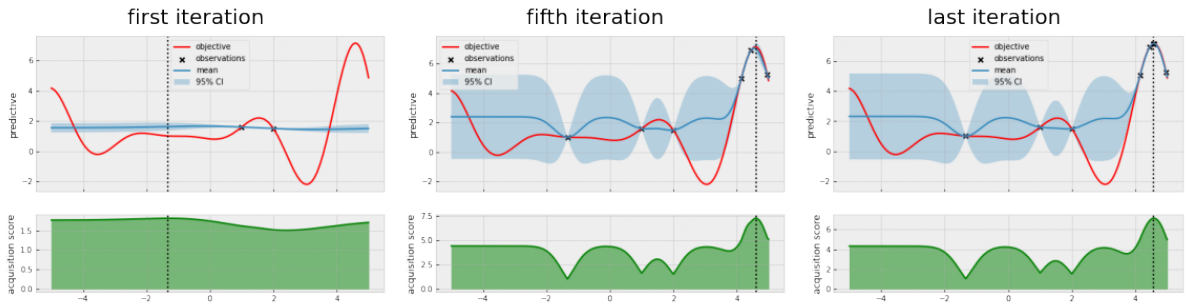


Figure 5.10 Progress made by the Upper Confidence Bound policy with the tradeoff parameter = 2. The policy successfully finds the global optimum.

Overall, the importance of using a good value for this tradeoff parameter while using the UCB policy is clear. However, again, it's difficult to say what value will work well for a given objective function. Exercise 1 of this chapter explores a certain strategy to adjust the value of this parameter as we progress throughout the search.

This marks the end of our discussion on UCB. We have seen that by adopting the "optimism in the face of uncertainty" mindset from the multi-armed bandit (MAB) problem, we obtain a BayesOpt policy whose explorative behavior can be directly controlled and tuned with a tradeoff parameter. In the next section, we will move on to the second policy that is taken from MAB, which has an entirely different motivation and strategy.

5.3 Smart sampling with the Thompson sampling policy

In this section, we learn about another heuristic in MAB that directly translates to a widely used BayesOpt policy called *Thompson sampling* (TS). As we will see, this policy uses an entire different motivation from that of UCB and therefore induces different optimization behavior. Similar to Section 5.2 with UCB, we will learn about the general idea behind this BayesOpt policy first and move on to its code implementation later.

5.3.1 One sample to represent the unknown

With UCB, we make decisions based on an optimistic estimate of the unknown quantity that we care about. This offers a simple way to reason about the consequences of the actions we take and the rewards we receive that trades off exploration and exploitation. What about TS?

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